

2) ①

• ALTERNATI VO =

1	3	4	6	8	12
2	5	7	9	10	11

sim ↙

1	4	12
2	5	10

NAO ↘

3	6	8
7	9	10

$$H(V) = - \sum_k P(V_k) \cdot \log_2 P(V_k)$$

$$H(V) = - \left[\frac{6}{12} \cdot \log_2 \left(\frac{6}{12} \right) + \frac{6}{12} \cdot \log_2 \left(\frac{6}{12} \right) \right]$$

$$H(V) = - \left[\frac{1}{2} \cdot -1 + \frac{1}{2} \cdot -1 \right]$$

$$H(V) = - [-0,5 + (-0,5)] = -[-1]$$

$$H(V) = 1 //$$

$$H(A) = \sum_{k=1}^d \frac{P_k + N_k}{P + N} H(E_k) \Rightarrow$$

$$H(A) = - \left[\left(\frac{6}{12} \cdot \left(\frac{3}{6} \cdot \log_2 \left(\frac{3}{6} \right) + \frac{3}{2} \cdot \log_2 \left(\frac{3}{6} \right) \right) + \left(\frac{6}{12} \cdot \left(\frac{3}{6} \cdot \log_2 \left(\frac{3}{6} \right) + \frac{3}{2} \cdot \log_2 \left(\frac{3}{2} \right) \right) \right) \right]$$

$$H(A) = - \left[\left(\frac{1}{2} \cdot \left(\frac{3}{6} \cdot -1 + \frac{3}{2} \cdot -1 \right) \right) + \left(\frac{1}{2} \cdot \left(\frac{3}{6} \cdot -1 + \frac{3}{2} \cdot -1 \right) \right) \right]$$

$$H(A) = - \left[\frac{1}{2} (-0,5 + 0,5) + \frac{1}{2} (-0,5 - 0,5) \right]$$

$$H(A) = -[-0,5 - 0,5]$$

$$H(A) = +1 //$$

$$IG(A) = H(V) - H(A)$$

$$IG(A) = 1 - 1$$

$$IG(A) = 0 //$$

-0,329

-0,459

S T O Q S S D

- CLIGNE: 4⁴⁵
- ALGUNS: 4²⁵
- CHUVA: 6⁴⁵
- NENHUM: 2⁰⁵

$$H(V) = - \left[\frac{6}{12} \log_2 \left(\frac{6}{12} \right) + \frac{6}{12} \log_2 \left(\frac{6}{12} \right) \right]$$

$$H(V) = 1 //$$

$$H(A) = - \left[\frac{4}{12} \left(\frac{4}{4} \log_2 \left(\frac{4}{4} \right) + \frac{0}{4} \log_2 \left(\frac{0}{4} \right) \right) + \frac{0}{12} \left(\frac{0}{2} \log_2 \left(\frac{0}{2} \right) + \frac{2}{2} \log_2 \left(\frac{2}{2} \right) \right) + \frac{6}{12} \left(\frac{2}{6} \log_2 \left(\frac{2}{6} \right) + \frac{4}{6} \log_2 \left(\frac{4}{6} \right) \right) \right]$$

$$H(A) = - [0 + 0 + (-0,459)]$$

$$IG(A) = 1 - 0,459$$

$$H(A) = 0,459 //$$

$$IG(A) = 0,541 //$$

- CHUVA: 3³⁵
- SIM: 5²⁵
- NAO: 7³⁵

$$H(V) = - \left[\frac{5}{12} \log_2 \left(\frac{5}{12} \right) + \frac{7}{12} \log_2 \left(\frac{7}{12} \right) \right]$$

$$H(V) = - [-0,979]$$

$$H(V) = +0,9798 \rightarrow 0,98$$

$$H(A) = - \left[\frac{5}{12} \left(\frac{3}{5} \log_2 \left(\frac{3}{5} \right) + \frac{2}{5} \log_2 \left(\frac{2}{5} \right) \right) + \frac{7}{12} \left(\frac{3}{7} \log_2 \left(\frac{3}{7} \right) + \frac{4}{7} \log_2 \left(\frac{4}{7} \right) \right) \right]$$

$$H(A) = - [-0,4045 + 0,5746]$$

$$IG(A) = 1 - 0,98$$

$$H(A) = - [-0,978]$$

$$IG(A) = 0,02 //$$

$$H(A) = +0,9781$$

$$IG(A) =$$

$$H(A) = - \left[\frac{3}{12} \left(\frac{1}{3} \log_2 \left(\frac{1}{3} \right) + \frac{2}{3} \log_2 \left(\frac{2}{3} \right) \right) + \frac{2}{12} \left(\frac{2}{2} \log_2 \left(\frac{2}{2} \right) + \frac{0}{2} \log_2 \left(\frac{0}{2} \right) \right) + \frac{7}{12} \left(\frac{3}{7} \log_2 \left(\frac{3}{7} \right) + \frac{4}{7} \log_2 \left(\frac{4}{7} \right) \right) \right]$$

$$H(A) =$$

$$H(A) = - [-0,2295 + 0 - 0,5746]$$

$$H(A) = +0,8041$$

$$IG(A) = 1 - 0,8041$$

$$IG(A) = 0,1959 \rightarrow 0,196 //$$

0,416
0,4709

data / /
(S) (T) (O) (Q) (S) (S) (D)

• BAR :

• Sim : 6 $\begin{matrix} \nearrow 3 \text{ sim} \\ \searrow 3 \text{ não} \end{matrix}$

• NÃO : 6 $\begin{matrix} \nearrow 3 \text{ sim} \\ \searrow 3 \text{ não} \end{matrix}$

$$IG(A) = 1 - 1 = 0 \quad \text{— mesmo cálculo da anterior}$$

• SEX SAB :

• Sim : 5 $\begin{matrix} \nearrow 2 S \\ \searrow 3 N \end{matrix}$

• NÃO : 7 $\begin{matrix} \nearrow 4 S \\ \searrow 3 N \end{matrix}$

$$H(A) = \left[\frac{5}{12} \cdot \log_2 \left(\frac{5}{12} \right) + \left(\frac{7}{12} \right) \cdot \log_2 \left(\frac{7}{12} \right) \right]$$

$$H(A) = \left[0,416 + 0,587 \cdot 0,78 \right]$$

$$H(A) = - \left[-0,52 + 0,457 \right]$$

$$H(A) = 0,9792$$

$$H(A) = \left[\frac{5}{12} \cdot \left(\frac{2}{5} \log_2 \left(\frac{2}{5} \right) + \frac{3}{5} \log_2 \left(\frac{3}{5} \right) \right) + \frac{7}{12} \cdot \left(\frac{4}{7} \log_2 \left(\frac{4}{7} \right) + \frac{3}{7} \log_2 \left(\frac{3}{7} \right) \right) \right]$$

$$H(A) = - \left[0,4046 + 0,5747 \cdot 0,99 + 0,587 \cdot 0,957 \cdot 0,8 + 0,43 \cdot 0,22 \right]$$

$$H(A) = 0,9792$$

$$H(A) = 0,9792$$

$$IG(A) = 1 - 0,9792$$

$$IG(A) = 0,0208 \quad \text{ou } 0,021$$

• TEMPO :

• 0-10 : 6 $\begin{matrix} \nearrow 4 S \\ \searrow 2 N \end{matrix}$

• 30-40 : 2 $\begin{matrix} \nearrow 1 S \\ \searrow 1 N \end{matrix}$

• >60 : 2 $\begin{matrix} \nearrow 0 S \\ \searrow 2 N \end{matrix}$

• 10-30 : 2 $\begin{matrix} \nearrow 1 S \\ \searrow 1 N \end{matrix}$

$$IG(A) = 1 - 0,7929$$

$$IG(A) = 0,2071$$

$$H(A) = - \left[\frac{6}{12} \left(\frac{4}{6} \log_2 \left(\frac{4}{6} \right) + \frac{2}{6} \log_2 \left(\frac{2}{6} \right) \right) + \frac{2}{12} \left(\frac{1}{2} \log_2 \left(\frac{1}{2} \right) + \frac{1}{2} \log_2 \left(\frac{1}{2} \right) \right) + \frac{2}{12} \left(\frac{0}{2} \log_2 \left(\frac{0}{2} \right) + \frac{2}{2} \log_2 \left(\frac{2}{2} \right) \right) + \frac{2}{12} \left(\frac{1}{2} \log_2 \left(\frac{1}{2} \right) + \frac{1}{2} \log_2 \left(\frac{1}{2} \right) \right) \right]$$

$$H(A) = - \left[-0,4591 - 0,3333 + 0 \right]$$

$$H(A) = - \left[-0,7924 \right]$$

$$H(A) = 0,7924$$

• FOMES

• SIM: 7 $\begin{matrix} 5S \\ 2N \end{matrix}$

$$H(A) = - \left[\frac{7}{12} \left(\frac{2}{7} \log_2 \left(\frac{2}{7} \right) + \frac{2}{7} \log_2 \left(\frac{2}{7} \right) \right) + \frac{5}{12} \left(\frac{4}{5} \log_2 \left(\frac{4}{5} \right) + \frac{1}{5} \log_2 \left(\frac{1}{5} \right) \right) \right]$$

$$H(A) = - [-0,5039 - 0,3008]$$

$$H(A) = 0,8042$$

$$IG(A) = 1 - 0,8042$$

$$IG(A) = 0,1958 \rightarrow 0,196$$

• RES

• SIM: 5 $\begin{matrix} 3S \\ 2N \end{matrix}$

• NAO: 7 $\begin{matrix} 3S \\ 4N \end{matrix}$

$$H(A) = - \left[\frac{5}{12} \left(\frac{3}{5} \log_2 \left(\frac{3}{5} \right) + \frac{2}{5} \log_2 \left(\frac{2}{5} \right) \right) + \frac{7}{12} \left(\frac{3}{7} \log_2 \left(\frac{3}{7} \right) + \frac{4}{7} \log_2 \left(\frac{4}{7} \right) \right) \right]$$

$$H(A) = - [-0,82 - 0,45]$$

$$H(A) = 0,9798$$

$$IG(A) = 1 - 0,979$$

$$IG(A) = 0,021$$

• TIPO

• FRANCES: 2 $\begin{matrix} 1S \\ 1N \end{matrix}$

• TAILANDES: 4 $\begin{matrix} 2S \\ 2N \end{matrix}$

• HAMBURGER: 4 $\begin{matrix} 2S \\ 2N \end{matrix}$

• ITALIANO: 2 $\begin{matrix} 1S \\ 1N \end{matrix}$

$$H(A) = - \left[\frac{2}{12} \left(\frac{1}{2} \log_2 \left(\frac{1}{2} \right) + \frac{1}{2} \log_2 \left(\frac{1}{2} \right) \right) + \frac{4}{12} \left(\frac{2}{4} \log_2 \left(\frac{2}{4} \right) + \frac{2}{4} \log_2 \left(\frac{2}{4} \right) \right) + \frac{4}{12} \left(\frac{2}{4} \log_2 \left(\frac{2}{4} \right) + \frac{2}{4} \log_2 \left(\frac{2}{4} \right) \right) + \frac{2}{12} \left(\frac{1}{2} \log_2 \left(\frac{1}{2} \right) + \frac{1}{2} \log_2 \left(\frac{1}{2} \right) \right) \right]$$

$$H(A) = - [-0,6666 - 0,3333 - 0,3333 - 0,6666]$$

$$H(A) = 1$$

$$IG(A) = 1 - 1 = 0$$

LISTA 2:

$$5) \text{PRECISION}_{\text{sim}} = \frac{15}{15+5} = \frac{15}{20} = 0,75$$

$$\text{PRECISION}_{\text{NÃO}} = \frac{10}{10+3} = \frac{10}{13} = 0,77$$

$$\text{RECALL}_{\text{sim}} = \frac{15}{10+3} = \frac{15}{13} = 0,83$$

$$\text{RECALL}_{\text{NÃO}} = \frac{10}{10+5} = \frac{10}{15} = 0,67$$

$$\text{F1-SCORE}_{\text{sim}} = \frac{2 * (0,83) * (0,75)}{0,83 + 0,75} = 0,787$$

$$\text{F1-SCORE}_{\text{NÃO}} = \frac{2 * (0,67) * (0,77)}{0,67 + 0,77} = 0,716$$

6)

$$\text{PRECISION}_A = \frac{10}{10+7} = \frac{10}{17} = 0,588$$

$$\text{PRECISION}_C = \frac{20}{20+6} = \frac{20}{26} = 0,769$$

$$\text{PRECISION}_B = \frac{15}{15+8} = \frac{15}{23} = 0,652$$

$$\text{PRECISION}_D = \frac{50}{50+6} = \frac{50}{56} = 0,892$$

$$\text{RECALL}_A = \frac{10}{10+7} = \frac{10}{17} = 0,588$$

$$\text{RECALL}_C = \frac{20}{20+10} = \frac{20}{30} = 0,666$$

$$\text{RECALL}_B = \frac{15}{15+3} = \frac{15}{18} = 0,833$$

$$\text{RECALL}_D = \frac{50}{50+7} = \frac{50}{57} = 0,877$$

$$\text{F1-SCORE}_A = \frac{2 * 0,588 * 0,588}{0,588 + 0,588} = 0,588$$

$$\text{F1-SCORE}_C = \frac{2 * 0,666 * 0,769}{0,666 + 0,769} = 0,713$$

$$\text{F1-SCORE}_B = \frac{2 * 0,833 * 0,652}{0,833 + 0,652} = 0,731$$

$$\text{F1-SCORE}_D = \frac{2 * 0,877 * 0,892}{0,877 + 0,892} = 0,884$$